

Computer Architectures

Exam of 31.8.2021 - part I

First name, Last name, ID.....

Question #1

Let consider the Branch Prediction Mechanism based on the Branch History Table (BHT).

You are requested to

1. Describe the hardware architecture of a BHT
2. Describe the behavior of a BHT, detailing when the processor accesses it, and which information it gets from it
3. Assuming that the processor uses 32 bit addresses, each instruction is 4 byte wide, and the BHT is composed of 8 entries, identify the final content of the BHT if
 - The BHT is initially empty (i.e., full of 0s, meaning NotTaken)
 - The following instructions are executed in sequence
 - `addi r2, r2, #1` located at the address `0x00B20020`
 - `add r3, r3, r2` located at the address `0x00B20024`
 - `bne r2, r3, l1` located at the address `0x00B20028`; the branch is taken, and the branch target address is `0x00A60050`
 - `addi r2, r2, #1` located at the address `0x00A60050`
 - `bez r2, l2` located at the address `0x00A60054`; the branch is not taken
 - `andi r5, r5, #1` located at the address `0x00A60058`
 - `bez r2, l3` located at the address `0x00A6005C`; the branch is taken, and the branch target address is `0x00BB0020`
 - `add r1, r2, r3` located at the address `0x00BB0020`
 - `bez r1, l4` located at the address `0x00BB0024`; the branch is taken, and the branch target address is `0x00A50050`.
4. For the example in the previous point, determine which of the 4 branch instructions were correctly predicted, and which were mispredicted.

0x00B20028	0010 1000	line 2	taken	mispredicted
0x00A60054	0101 0100	line 5	not taken	correctly predicted
0x00A6005C	0101 1100	line 7	taken	mispredicted
0x00BB0024	0010 0100	line 1	taken	mispredicted

Final content:

Line 0 0
Line 1 1
Line 2 1
Line 3 0
Line 4 0
Line 5 0
Line 6 0
Line 7 1

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Question #2

Let consider a MIPS64 architecture including the following functional units (for each unit the number of clock periods to complete one instruction is reported):

- Integer ALU: 1 clock period
- Data memory: 1 clock period
- FP arithmetic unit: 2 clock periods (pipelined)
- FP multiplier unit: 5 clock periods (pipelined)
- FP divider unit: 8 clock periods (unpipelined)

You should also assume that

- The branch delay slot corresponds to 1 clock cycle, and the branch delay slot is not enabled
- Data forwarding is enabled
- The EXE phase can be completed out-of-order.

You should consider the following code fragment and, filling the following tables, determine the pipeline behavior in each clock period, as well as the total number of clock periods required to execute the fragment. The values of the constants k1 and k2 are written in f10 and f11 before the beginning of the code fragment.

```

; ***** MIPS64 *****
;  for (i = 0; i < 10; i++) {
;
;      v4[i] = ((v1[i]*v2[i]/k1))+v3[i]/k2;
;  }

```

```

      .data
v1:    .double "10 values"
v2:    .double "10 values"
v3:    .double "10 values"
v4:    .double "10 values"

```

```

        .text
main:    daddui r1,r0,0
        daddui r2,r0,10
loop:    l.d  f1,v1(r1)
        l.d  f2,v2(r1)
        mul.d f3, f1, f2
        div.d f4, f3, f10
        l.d  f5,v3(r1)
        div.d f6, f5, f11
        add.d f7,f4,f6
        s.d  f7,v4(r1)
        daddui r1,r1,8
        daddi r2,r2,-1
        bnez r2,loop
        halt

```

Comments	Clock cycles
$r1 \leftarrow \text{pointer}$	
$r2 \leq 10$	
$f1 \leq v1[i]$	
$f2 \leq v2[i]$	
$f3 \leq v1[i] * v2[i]$	
$f4 \leq v1[i] * v2[i] / k1$	
$f5 \leq v3[i]$	
$f6 \leq v3[i] / k2$	
$f7 \leq v1[i] * v2[i] / k1 + v3[i] / k2$	
$v4[i] \leq f7$	
$r1 \leq r1 + 8$	
$r2 \leq r2 - 1$	

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Exam of 18.6.2020 - part I

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[illegible]